

# Richard Hallett

Full-stack and AI engineer building systems that fail closed.  
Before this came fifteen years as a therapist, and the same instinct now points at machines.

kai@oceanheart.ai · [github.com/rickhallett](https://github.com/rickhallett) · [oceanheart.ai](https://oceanheart.ai)



|                                                 |                                                        |                                                                     |
|-------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------|
| BASED<br>Dorset, UK<br>Will consider relocating | ENGINEERING<br>Enterprise: 5 years<br>Agentic: 2 years | PRIOR CAREER<br>Cognitive Behavioural<br>Psychotherapist since 2004 |
|-------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------|

The last five years have gone into shipping production TypeScript, Python and Go, most lately designing AI systems that fail closed. These systems use deterministic validators that gate LLM output before it reaches a customer, keep every turn audit-traced, and track the cost of each run. Fifteen years as a CBT therapist carry over in practice and habit: spotting plausible-but-wrong output and asking better questions whilst a system remains uncertain.

## — SELECTED WORK

### LoanSlam [live demo ↗](#) [live site ↗](#)

Fail-closed conversation engine for regulated lending. The LLM proposes each turn whilst a deterministic validator enforces compliance and blocks unsafe output before it can reach a customer; every turn is audit-traced. Drivable demo and an evaluation-reports page, both over synthetic data.

Vue · TypeScript · Node · Zod

### The Pit [thepit.cloud ↗](https://thepit.cloud)

Evaluation platform for multi-agent systems, with observable traces, rubric scoring, failure tagging, and cost tracking per run. It answers the commercial questions behind AI tooling: what did this run cost, what did it catch, and is the extra model call worth it?

Next.js · TypeScript · Postgres · Stripe

### Sortie [/rickhallett/sortie ↗](https://rickhallett/sortie)

AI-assisted code review. Three LLMs review a diff in parallel, a fourth synthesises, and merge is gated on convergent severity.

Multi-model review · convergent gating

### Halo [/rickhallett/haLo ↗](https://rickhallett/haLo)

Personal engineering tool-layer: CLI modules for work tracking, memory, briefing synthesis, email triage, journaling, and agent orchestration. Each module has an explicit CLI surface and isolated storage, so humans and agents use the same tools.

Python · SQLite · NATS JetStream · Docker · Kubernetes

### Becoming Diamond client product

Shipped client product: marketing site and gated member portal delivering a 30-day video course, with AI chat, Stripe payments, and a git-based CMS for client editing. Built across front end, backend, payments, content workflows, and deployment.

Next.js · React 19 · TypeScript · Stripe · Decap CMS

## — EXPERIENCE

2024 to present · Oceanheart.ai · independent

**Software Engineer.** Client-facing full-stack work and AI delivery tooling, covering scoping, build planning, implementation, integration, deployment and support.

2026 (May to June) · Loans by MAL · contract

**Full-Stack Developer.** Built the first fail-closed AI conversation engine for a UK lending context, with a Vue/TS front end, a Node/TS backend, shared Zod contracts, deterministic safety validation, and persona-driven simulation of confused, adversarial and vulnerable journeys.

2023 to 2024 · EDITED · retail analytics

**Software Engineer.** Data-visualisation features and enterprise API integrations for an AI-driven retail analytics SaaS. React · TypeScript · Python.

2021 to 2023 · Brandwatch · social intelligence

**Software Engineer.** Greenfield enterprise data-viz platform in a team of five; scalable React components, modernised legacy Backbone, mentored junior developers.

2020 to 2021 · Telesoft

**Full-Stack Engineer.** Secure features for cybersecurity apps. TS · Angular · Node.

2019 to 2020 · School Business Services

**Software Developer.** Vue front end for school financial-management software.

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2004 to present · NHS / Private Practice

**Cognitive Behavioural Therapist.** Fifteen years of clinical practice, NHS and private; now a small private caseload alongside engineering. The other half of the résumé, kept deliberately short.

## — EDUCATION & TECHNICAL

### EDUCATION

**PGDip** Cognitive Behavioural Therapy  
Royal Holloway, University of London

**PGCert** Primary Mental Healthcare  
University of Central Lancashire

**BSc** Psychology  
UWE Bristol

### TECHNICAL

LANG TypeScript, Python, Go, Bash

FE React, Next.js, Vue, Angular, Astro, Tailwind

BE Node, FastAPI, REST, PostgreSQL, SQLite, NATS

OPS Git, GitHub Actions, Vercel, Docker, AWS, Terraform/OpenTofu, Kubernetes

AI LLM integration, agent/tool-layer architecture, provider-agnostic adapters, AI-assisted review, rubric evaluation, cost/token tracking, fail-closed safety patterns

## — INTERESTS

Strength and kettlebell training. Meditation. Ecstatic dance, electronic music and hard rock. Authentic relating and ongoing self-development work. In the distant past, a champion bodybuilder.

The same thing runs through all of it: pick a hard practice, repeat it until it changes you. Engineering included.